Analysis and Numerical Exploration of a PDE-Based Model for Nitrifying Bacteria

Thesis report

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Chapter 1

Introduction

This thesis looks at a mathematical model of nitrifying bacteria, the bacteria that remove ammonium from wastewater. These bacteria live together in dense clumps, and the model describes how they grow, move and use food. The model is a system of seven coupled partial differential equations. It builds of the work of Dr. Viktoria Freingruber, Dr. Rebeca Gonzalez-Cabaleiro and Dr. Havva Yoldaş.

We have two goals. The first is to understand the model better, using mathematics. The second is to build a computer program, called a solver, that can run the model with realistic parameters.

This chapter explains why the problem matters and what we set out to do. The model itself is described in the next chapter.

1.1 Why nitrification matters

Ammonium (NH_4^+) is a common pollutant in wastewater. If it reaches a river or a lake, it feeds algae and uses up the oxygen that fish and other life need. Treatment plants therefore turn ammonium into less harmful substances.

The first step of this is called *nitrification*. Bacteria use oxygen to convert ammonium into nitrite (NO_2^-), and then into nitrate (NO_3^-).

In treatment plants, these bacteria rarely float around on their own. They live packed together in dense clumps. A *biofilm* is a layer of bacteria growing on a surface. A *granule* is a small ball of bacteria floating in the water. We call both of them *aggregates*.

Space matters inside an aggregate. The bacteria need dissolved substances such as ammonium and oxygen. We call these *substrates*. They come from the surrounding water and have to travel into the clump by diffusion. The cells near the surface use them first, so the inside gets very little. A granule can therefore have a well-supplied outer layer and a starved core with no oxygen. Different species do best at different depths, so the community sorts itself into layers. This affects how well the community removes nitrogen. A model of a well-mixed tank cannot show this, because it has no notion of position.

Experiments on aggregates are slow and expensive. Mathematical models help here. They let us test ideas on a computer, plan experiments, and predict long-term behaviour from a few key properties, such as growth rates and how the species interact.

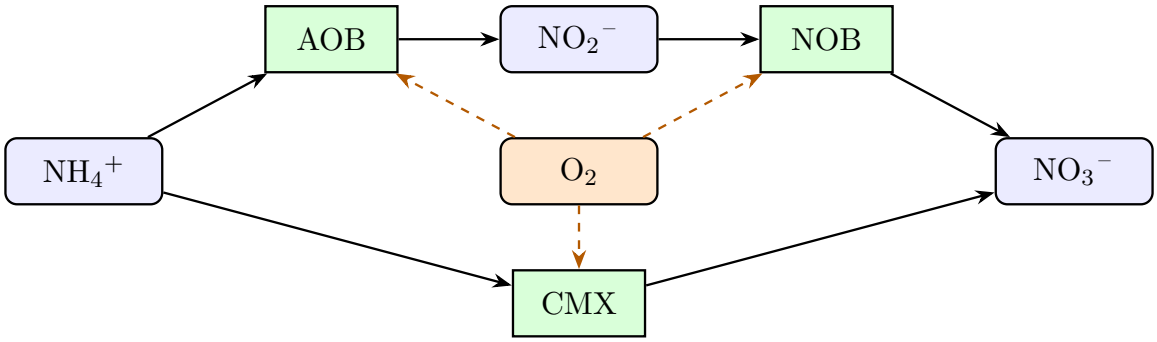


Figure 1.1: The two routes of nitrification. AOB and NOB work one after the other (top route). CMX does both steps alone (bottom route). Solid arrows show the flow of nitrogen. Dashed arrows show that all three groups need oxygen.

1.2 Three kinds of nitrifying bacteria

Ammonia-oxidising bacteria (AOB) turn ammonium into nitrite. *Nitrite-oxidising bacteria* (NOB) turn nitrite into nitrate. *Comammox* bacteria (CMX), short for "complete ammonia oxidation", do both steps on their own [1, 10].

Each group carries out one chemical reaction:

$$\begin{aligned}
&\mathrm{NH_4^+ + 1.5\,O_2 \longrightarrow NO_2^- + 2\,H^+ + H_2O} && \text{(AOB)},\\
&\mathrm{NO_2^- + 0.5\,O_2 \longrightarrow NO_3^-} && \text{(NOB)},\\
&\mathrm{NH_4^+ + 2\,O_2 \longrightarrow NO_3^- + 2\,H^+ + H_2O} && \text{(CMX)}.
\end{aligned}$$

All three reactions use oxygen. Figure 1.1 shows how the three groups are connected.

The three groups affect each other in three ways:

- **AOB and NOB (commensalism):** NOB lives on the nitrite that AOB makes. This is one-way help: NOB benefits and AOB is not affected.
- **AOB and CMX: competition for ammonium.** Both eat ammonium, so what one takes is no longer available to the other.
- **All three: competition for oxygen.** Every group needs oxygen, so they all compete for it.

This mix of helping and competing is what makes the system interesting. Which group wins, and where, depends on how much ammonium and oxygen reaches each spot in the aggregate.

1.3 Ways to model microbial communities

There are three main approaches.

ODE models describe average concentrations and densities in a well-mixed reactor. They are simple and well understood. They have no space, so they cannot describe diffusion limits or the layers inside an aggregate.

Individual-based models (IBMs) follow every cell: its position, size, growth and movement. The cells are coupled to a diffusion equation for the substrates [4]. IBMs have been used for aggregates and biofilms [6–8]. They capture microscopic detail and reproduce spatial patterns. They are expensive to run and hard to analyse mathematically.

Continuum PDE models describe densities that vary smoothly in space and time [11]. They are cheaper to simulate, and PDE analysis can be applied to them. The challenge is to build a continuum model that keeps the main behaviour of the individual-based one.

1.4 The source model and the question of this thesis

The source paper [3] proposes a continuum model motivated by the individual-based model of Martinez-Rabert et al. [7]. It tracks the density of each bacterial species and the concentration of each substrate in space and time. Bacteria grow when substrates are available, die faster when crowded, and are pushed away from crowded regions. Substrates diffuse, and the bacteria consume or produce them. For n species and m substrates the model has $n + m$ coupled PDEs.

The authors test the model on three known results from individual-based studies:

- three species competing for one substrate;
- a fast-growing species against an efficient species;
- a chain of three species that feed each other.

They find the same qualitative patterns. They report that a simulation takes seconds, compared with hours for the individual-based model. They also derive the model from a random walk on a lattice, and they prove that a symmetric version of the competition model is well posed for short times.

The nitrification system (three species, four substrates, seven PDEs) is written down in the paper but not analysed. The authors note that, for realistic parameters, the processes run on very different time scales, and they leave the study for future work.

This difference in time scales is the central difficulty. Substrates diffuse across an aggregate quickly, while bacteria grow and move slowly. A method that follows the substrates in time needs tiny steps, although the bacteria hardly change during them. This is wasteful.

A simplification is to let the substrates adjust instantly to the current bacterial densities. The time derivative then disappears from the substrate equations, and they become *elliptic* equations that are solved again whenever the bacterial densities change. The result is an *elliptic–parabolic* system. This is called a *fast–slow* or *quasi-steady-state* reduction. Whether it is justified is the subject of Tikhonov's theorem and related results of singular perturbation theory [2, 5, 9].

The questions of this thesis are therefore:

- Is the reduction to an elliptic–parabolic system justified for the nitrification model?
- Which mathematical properties does the reduced model have?
- How can it be solved reliably and efficiently, in one and two space dimensions, with realistic parameters?

1.5 Objectives

The project description from the supervisors lists the following goals:

1. Understand the nitrification process and the PDE model of [3].
2. Analyse the fast–slow time-scale separation mathematically.
3. Adapt the numerical simulations to the nitrification system of Figure 1.1 with realistic parameters.
4. Improve the efficiency of the numerical solver.
5. Analyse the elliptic–parabolic system mathematically.
6. Use the solver to study how the availability of ammonium and oxygen affects the competition between AOB and comammox bacteria in a spatially growing community.

1.6 Contributions and what is not new

From the source paper. The model, its special cases for competition and commensalism, the three verification scenarios, the derivation from a lattice model, the local well-posedness result for the symmetric competition case, and the travelling-wave analysis are due to [3]. We review them in this thesis so that it can be read on its own. The parameter values come from the supervisors' research group and from the cited literature. Our own work on the parameters is the conversion of units, the derivation of missing values, and the documentation of the assumptions made.

New in this thesis.

- A dimensionless form of the seven-PDE system with an explicit small parameter, and a dictionary between the original and the dimensionless variables.
- A mathematical analysis of the elliptic–parabolic system: conservation laws of the reaction network, positivity and bounds, the boundary conditions for which a quasi-steady state exists at all, and the validity of the reduction. Each result is marked as proved, as depending on an assumption, or as checked only numerically. Parts of the algebra were also verified with the proof assistant Lean 4.
- A numerical solver for the reduced system in one dimension (slab, cylinder, sphere) and in two dimensions, together with a second solver that keeps the time derivative of the substrates and is used to test the reduction.
- Verification of the solver against closed-form solutions, and validation against the results of the source paper.
- Numerical experiments, including pattern formation under different boundary conditions and the competition between AOB and comammox bacteria.

1.7 Outline of the thesis

The thesis has four parts.

Part I: Model and setting. Chapter 2 presents the source model, including the seven-PDE nitrification system, and introduces fast–slow systems. Chapter 3 collects all parameter sets used. Chapter 4 describes the scenarios and figures of the source paper, which later serve as targets for validation. Chapter 5 makes the equations dimensionless and gives the dictionary between the original and the dimensionless form.

Part II: Analysis. Chapter 6 studies the reduced elliptic–parabolic system mathematically.

Part III: Numerics. Chapter 7 describes the architecture of the solver, with all numerical methods and approximations. Chapter 8 reports the tests. Chapter 9 presents the numerical experiments.

Part IV: Closing. Chapter 10 summarises the results, discusses their limits and lists open questions. The appendices contain the full parameter tables, derivations, longer proofs, a guide to the code and additional figures.

Chapter 2

The Source Model

This chapter presents the source model [3]. It defines the setting, writes down the seven PDEs of the nitrification system and explains every term. It ends with an introduction to fast–slow systems.

2.1 The biology and the chemical reactions

The system has three bacterial species and four substrates. The species are AOB, NOB and CMX. The substrates are ammonium (NH_4^+), nitrite (NO_2^-), nitrate (NO_3^-) and oxygen (O_2).

Table 2.1 shows what each species consumes and produces. Every species needs two substrates at the same time: a nitrogen substrate (ammonium or nitrite) and oxygen. If either one is missing, the species cannot grow.

The substrates also differ in where they come from:

- **Ammonium and oxygen** are supplied from outside the aggregate, through its boundary.
- **Nitrite and nitrate** are not supplied from outside. They exist only as products of the bacteria. Nitrite is used up again by NOB. Nitrate is never consumed.

The reactions of Section 1.2 fix how much oxygen each species uses per nitrogen atom it processes:

	AOB	NOB	CMX
Oxygen used per nitrogen atom	1.5	0.5	2

CMX uses the sum of the AOB and NOB amounts, because it performs both steps. Each reaction moves one nitrogen atom from the consumed substrate to the product, so nitrogen is conserved.

Two further quantities link the bacteria to the substrates in the model:

- the *yield* Y, the amount of biomass made per unit of substrate consumed;
- a *conversion factor* σ that sets how much product is released per unit of substrate consumed.

Their values are given in Chapter 3, and the conversion factors are written out in Section 2.5.

Species	Consumes	Produces
AOB	NH_4^+, O_2	NO_2^-
NOB	NO_2^-, O_2	NO_3^-
CMX	NH_4^+, O_2	NO_3^-

Table 2.1: What each species consumes and produces.

2.2 Modelling assumptions

The model rests on four assumptions about how bacteria move, grow and die, and how substrates behave.

Movement of the bacteria. Bacteria move by two mechanisms: random motion, which acts like diffusion, and being pushed down the gradient of the population pressure. In dense aggregates the pushing dominates, because growing cells shove their neighbours aside. We therefore take the random part to be much smaller than the pushing part. The pressure grows linearly with the total biomass density $\rho = u_1 + u_2 + u_3$. All three species feel the same pressure, so each species is pushed by the gradient of the *total* density, not only of its own.

Growth. Growth follows Monod kinetics. For a substrate with concentration c and half-saturation constant K, the Monod function is

$$M(c; K) = \frac{c}{K + c}.$$

It is 0 when the substrate is absent, equals $\frac{1}{2}$ when $c = K$, and approaches 1 when the substrate is plentiful. A species that needs two substrates grows at a rate proportional to the product of two Monod functions, so it stops growing if either substrate is missing.

Death. The death rate of a species grows with the total density ρ. This represents crowding and competition for resources. It limits the population: with plentiful substrates, growth and death balance when $\rho = r_i/b_i$, where r_i is the maximal growth rate and b_i the death coefficient of species i.

Substrates. Substrates spread by ordinary (Fickian) diffusion. The bacteria consume them with Monod kinetics, scaled by the yield, and release products as by-products. At the boundary of the aggregate the substrates can be supplied in three ways: with a prescribed concentration (Dirichlet condition), a prescribed flux (Neumann condition), or a combination of the two. The paper uses Neumann conditions. Other choices are studied later in this thesis.

2.3 The general framework: n species and m substrates

The paper writes the model for any number n of species and m of substrates, in a domain $\Omega \subset \mathbb{R}^2$ [3]. Let $u_i(x, t)$ be the density of species $i = 1, \ldots, n$, let $c_j(x, t)$ be the concentration of

substrate $j = 1, \ldots, m$, and let $\rho = \sum_i u_i$ be the total density. The model is

$$\partial_t u_i = \nabla \cdot [d_i \nabla u_i + a_i u_i \nabla f(\rho)] + u_i \Big(r_i \prod_{j=1}^{m} \frac{c_j}{K_{ij} + c_j} - b_i \rho \Big), \tag{2.1}$$

$$\partial_t c_j = D_j \Delta c_j - \sum_{i=1}^{n} u_i \, \delta_{ij} \, \frac{r_i}{Y_{ij}} \prod_{k=1}^{m} \frac{c_k}{K_{ik} + c_k} + \sum_{i=1}^{n} \sigma_{ij} \, u_i \, r_i \prod_{\substack{k=1 \\ k \neq j}}^{m} \frac{c_k}{K_{ik} + c_k}. \tag{2.2}$$

The first equation is for the bacteria and the second is for the substrates. For the nitrification system, all seven equations are written out explicitly, with species and substrate names instead of indices, in Chapter A.

The bacterial equation. The terms on the right of Equation (2.1) are, in order:

- $d_i \nabla u_i$: random motion of species i, with coefficient d_i;
- $a_i u_i \nabla f(\rho)$: movement down the pressure gradient, with coefficient a_i;
- $r_i u_i \prod_j c_j/(K_{ij} + c_j)$: growth, with maximal rate r_i and a Monod factor for each substrate;
- $-b_i u_i \rho$: death, which grows with the total density.

The substrate equation. The terms on the right of Equation (2.2) are, in order:

- $D_j \Delta c_j$: diffusion of substrate j, with coefficient D_j;
- the first sum: consumption by the bacteria, equal to the growth rate divided by the yield Y_{ij};
- the second sum: production of substrate j by the species that make it, with conversion factor σ_{ij}.

Which species uses which substrate. Two sets of indicators decide which terms are active:

- If species i consumes substrate j, then $\delta_{ij} = 1$ and the constants $K_{ij} > 0$ and $Y_{ij} > 0$ are given.
- Otherwise $\delta_{ij} = 0$ and $K_{ij} = 0$. The factor $c_j/(K_{ij} + c_j)$ then equals 1 wherever $c_j > 0$, so substrate j does not limit the growth of species i. The yield Y_{ij} is not needed.
- If substrate j is a product of species i, then $\sigma_{ij} > 0$. Otherwise $\sigma_{ij} = 0$.
- A species does not both consume and produce the same substrate.

The pressure. The function f is the internal pressure. It is smooth and increasing, with $f(0) = 0$. We take $f(\rho) = \rho$ throughout. If crowding is the main driver of movement, then $d_i \ll a_i$.

Initial and boundary conditions. The model is completed by

$$\begin{aligned} u_i(x,0) = u_i^0(x) \geq 0, \qquad & c_j(x,0) = c_j^0 \geq 0 && \text{in } \Omega, \\ \partial_\nu u_i = 0, \qquad & \partial_\nu c_j = \gamma_j \geq 0 && \text{on } \partial\Omega, \end{aligned}$$

where ν is the outward unit normal and c_j^0 and γ_j are constants. The bacteria cannot leave the domain. The constant γ_j is a constant supply of substrate j through the boundary; it is zero for substrates that are not supplied from outside.

2.4 Two special cases: competition and commensalism

The paper studies two simpler versions of the general model [3]. They are used in Chapter 4 and Chapter 8 to check our solver against the paper.

2.4.1 Competition for one substrate

Here $n \geq 2$ species compete for a single substrate with concentration c:

$$\partial_t u_i = \nabla \cdot [d_i \nabla u_i + a_i u_i \nabla \rho] + u_i \Big(r_i \frac{c}{K_i + c} - b_i \rho \Big), \qquad i = 1, \ldots, n, \tag{2.3}$$

$$\partial_t c = D \Delta c - \sum_{i=1}^{n} u_i \, \frac{r_i}{Y_i} \, \frac{c}{K_i + c}. \tag{2.4}$$

The initial and boundary conditions are $u_i(x,0) = u_i^0(x) \geq 0$, $c(x,0) = c^0 > 0$, $\partial_\nu u_i = 0$ and $\partial_\nu c = \gamma > 0$.

Species that use the same substrate cannot all survive in the same place: the better competitor excludes the others. Individual-based models nevertheless show that the species separate into radial sectors, which allows less competitive species to persist. This is called *columned stratification*. The PDE model reproduces it (Chapter 4).

2.4.2 Commensalism along a chain

Here three species and three substrates form a chain:

- species 1 consumes substrate 1, which is supplied from outside, and produces substrate 2;
- species 2 consumes substrate 2 and produces substrate 3;
- species 3 consumes substrate 3.

The equations are

$$\partial_t u_i = \nabla \cdot [d_i \nabla u_i + a_i u_i \nabla \rho] + u_i \Big(r_i \frac{c_i}{K_i + c_i} - b_i \rho \Big), \qquad i = 1, 2, 3, \tag{2.5}$$

$$\partial_t c_1 = D_1 \Delta c_1 - u_1 \frac{r_1}{Y_1} \frac{c_1}{K_1 + c_1}, \tag{2.6}$$

$$\partial_t c_2 = D_2 \Delta c_2 - u_2 \frac{r_2}{Y_2} \frac{c_2}{K_2 + c_2} + \sigma_{12} \, u_1 \frac{r_1}{Y_1} \frac{c_1}{K_1 + c_1}, \tag{2.7}$$

$$\partial_t c_3 = D_3 \Delta c_3 - u_3 \frac{r_3}{Y_3} \frac{c_3}{K_3 + c_3} + \sigma_{23} \, u_2 \frac{r_2}{Y_2} \frac{c_2}{K_2 + c_2}. \tag{2.8}$$

Unknown	Increases through	Decreases through
u_{AOB}	growth on $NH_4{}^+$ and O_2	crowding-related death
u_{NOB}	growth on $NO_2{}^-$ and O_2	crowding-related death
u_{CMX}	growth on $NH_4{}^+$ and O_2	crowding-related death
$c_{\mathrm{NH_4}}$	supply through the boundary	consumption by AOB and CMX
$c_{\mathrm{NO_2}}$	production by AOB	consumption by NOB
$c_{\mathrm{NO_3}}$	production by NOB and CMX	nothing (it is never consumed)
$c_{\mathrm{O_2}}$	supply through the boundary	consumption by AOB, NOB and CMX

Table 2.2: The seven equations of the nitrification system.

Here σ_{12} and σ_{23} are the conversion factors from substrate 1 to 2 and from 2 to 3. Only substrate 1 is present at the start and supplied through the boundary, with $c_1(x,0) = c_1^0 > 0$ and $\partial_\nu c_1 = \gamma_1 > 0$. The substrates 2 and 3 start at zero and have zero flux, $\partial_\nu c_2 = \partial_\nu c_3 = 0$.

Individual-based models show *layered stratification* in this case: the species form concentric layers, ordered by the position of their substrate in the chain. The PDE model reproduces this as well (Chapter 4).

2.5 The nitrification system: seven PDEs

The nitrification system is the general model of Section 2.3 with $n = 3$ species and $m = 4$ substrates. The species are AOB, NOB and CMX, and the substrates are $NH_4{}^+$, $NO_2{}^-$, $NO_3{}^-$ and O_2. This gives three equations for the bacteria and four for the substrates, seven in total. Which species uses and produces which substrate is given in Table 2.1; this fixes the indicators δ_{ij} and σ_{ij} of the general model. All seven equations are written out with names in Chapter A.

Table 2.2 summarises what each equation contains. In addition, every equation has a transport term: random motion and pressure-driven movement for the bacteria, diffusion for the substrates.

Production terms. The paper leaves the conversion factors σ_{12}, σ_{23} and σ_{13} (nitrite from AOB, nitrate from NOB, nitrate from CMX) as free positive constants. In our solver, the amount of product released equals the amount of the nitrogen substrate that the species consumes. Each nitrogen atom taken up is released as one nitrogen atom in the product, so nitrogen is conserved. In the notation of the paper this means $\sigma = 1/Y$ with Y the yield on the nitrogen substrate, for example $\sigma_{12} = 1/Y_{\mathrm{AOB,NH_4}}$.

Initial and boundary conditions. The paper uses zero flux for the three bacterial species. Ammonium and oxygen are supplied through the boundary with a constant flux. Nitrite and nitrate are only produced by the bacteria and have zero flux. Initially, ammonium and oxygen are present and nitrite and nitrate are absent. The explicit conditions are in Chapter A. This thesis also uses Dirichlet and Robin conditions for the substrates (Chapter 5).

What the paper does with this system. The paper writes the system down but does not simulate or analyse it. It notes that the realistic parameters involve very different time scales,

and leaves the system for future work.

2.7 Multiple time scales and fast–slow systems

2.7.1 Time scales in the model

The model has two very different time scales. A substrate crosses an aggregate of size L by diffusion in the time $\tau_{\text{fast}} = L^2/D$. The bacteria grow on the time scale $\tau_{\text{slow}} = 1/r_{\max}$, where $r_{\max}$ is the largest growth rate. Their movement is slower still. The ratio of the two is

$$\varepsilon = \frac{\tau_{\text{fast}}}{\tau_{\text{slow}}} = \frac{r_{\max} L^2}{D}. \tag{2.9}$$

For the parameter sets of this thesis, ε lies between 10^{-3} and 10^{-2} (Chapter 5). Substrates therefore reach their balance with the bacteria at least a hundred times faster than the bacteria change.

This is a problem for numerical methods. A method that treats the diffusion explicitly must take time steps set by the diffusion time across one grid cell, which is even shorter than τ_{fast}. It must take many such steps to follow the bacteria over a time τ_{slow}. Implicit methods avoid the step restriction, but they still solve the fast substrate problem at every step.

2.7.2 Slow manifolds and Tikhonov's theorem

The standard setting is a system of ordinary differential equations with a slow variable x and a fast variable y:

$$x' = f(x, y), \qquad \varepsilon\, y' = g(x, y), \tag{2.10}$$

with a small parameter $\varepsilon > 0$. Setting $\varepsilon = 0$ turns the second equation into the algebraic condition $g(x, y) = 0$. If it can be solved as $y = h(x)$, the set $\{y = h(x)\}$ is called the *slow manifold*, and the *reduced system* is

$$x' = f(x, h(x)).$$

Tikhonov's theorem [9] says when the reduced system is a good approximation. Suppose that for each x the root $y = h(x)$ is isolated and attracts the fast dynamics $\mathrm{d}y/\mathrm{d}\tau = g(x, y)$, that is, $\partial_y g(x, h(x))$ has eigenvalues with negative real part. Suppose also that the initial value $y(0)$ lies in the basin of attraction of $h(x(0))$. Then, as $\varepsilon \to 0$, the solution of Equation (2.10) converges to the solution of the reduced system on any fixed time interval. The only exception is a short initial layer, of length of order ε, in which y moves quickly onto the slow manifold.

Example 2.1. Take $x' = -y$ and $\varepsilon y' = x - y$. Then $g(x, y) = x - y = 0$ gives $y = h(x) = x$, and $\partial_y g = -1 < 0$, so the root is stable. The reduced equation is $x' = -x$. The full solution follows $y \approx x$ after a layer of length of order ε.

There is also a geometric version of the theory. Fenichel [2] shows that for small ε the slow manifold persists as an invariant manifold that the solutions approach. Henry [5] extends these ideas to parabolic PDEs. In our case the fast part is a reaction–diffusion problem for the

substrates, and the stability condition becomes a condition on the spectrum of its linearisation. Chapter 6 states and uses this version.

2.7.3 The quasi-steady-state reduction

We apply this idea to the substrate equations. Write the reaction part of Equation (2.2) as $R_j(u, c)$, so that

$$\partial_t c_j = D_j \Delta c_j + R_j(u, c), \qquad j = 1, \dots, m. \tag{2.11}$$

On the slow time scale, with lengths measured in units of L, this equation has the form

$$\varepsilon\, \partial_t c_j = \Delta c_j + R_j(u, c),$$

with the diffusion coefficient absorbed into the scaling (Chapter 5). Setting $\varepsilon = 0$ removes the time derivative:

$$0 = D_j \Delta c_j + R_j(u, c). \tag{2.12}$$

This is called the *quasi-steady-state* reduction. The substrates are assumed to be in balance with the current bacterial densities u at every moment.

What is kept is the slow behaviour of the bacteria. What is lost is the initial layer: if the initial substrate concentrations do not satisfy Equation (2.12), the true solution adjusts within a time of order ε, and the reduced model does not describe this adjustment.

2.7.4 The elliptic–parabolic system

Combining the two parts gives the reduced model. For each time t and given densities u, the substrate concentrations $c = c^*(u)$ solve the elliptic problem Equation (2.12) with the boundary conditions. The bacteria then follow

$$\partial_t u_i = \nabla \cdot \left[d_i \nabla u_i + a_i u_i \nabla \rho \right] + u_i \Big(r_i \prod_{j=1}^{m} \frac{c_j^*(u)}{K_{ij} + c_j^*(u)} - b_i \rho \Big). \tag{2.13}$$

The bacterial equations are parabolic, because they contain a time derivative. The substrate equations are elliptic, because they do not. The system is therefore called *elliptic–parabolic*.

For the reduced model to make sense, the elliptic problem must have exactly one solution $c^*(u)$ for every u that occurs. Whether this holds depends on the boundary conditions. For the quasi-steady state to be a good approximation, $c^*(u)$ must also be stable for the fast dynamics, as in Tikhonov's theorem. Chapter 6 shows when these conditions hold.

Chapter 3

Parameters

This chapter collects all parameter sets used in the thesis. For each set it gives the source of the values, the units, and the assumptions made when the data were cleaned. The complete tables are collected in the appendices.

3.1 Overview and units

3.1.1 The parameter sets

Three kinds of parameter sets are used:

- **Presets.** Three dimensional sets of the full nitrification system, stored in the code under the names `toy`, `rebeca` and `eloi`.
- **Paper sets.** The dimensionless parameters of the source paper for its three verification scenarios (Cases A, B and C) [3].
- **Simulation sets.** The values actually used in each numerical experiment. They combine a preset or a paper set with choices for the boundary conditions, the initial biomass and the numerical settings.

3.1.2 Units

All dimensional parameters are stored in the same units:

Quantity	Unit
growth and death rates r_i, b_i	1/day
half-saturation constants K_{ij} and concentrations c_j	μM
diffusion coefficients d_i, D_j	m^2/day
domain size L	μm
yields Y_{ij}	mol/mol

Values from the sources are converted to these units as follows:

- rates given per hour are multiplied by 24;

Set	Origin	Used for
`toy`	round numbers chosen for testing	solver development and tests
`rebeca`	rough estimate from R. Gonzalez-Cabaleiro	nitrification simulations
`eloi`	kinetics derived from thermodynamics [7, 8], literature diffusivities	nitrification simulations
Paper Cases A, B, C	Table 1 of the source paper	validation against the paper

Table 3.1: The parameter sets and what they are used for.

- diffusion coefficients given in m^2/s are multiplied by 86 400, and those given in m^2/h by 24;
- concentrations given in mol/L are converted to μM, and those given in mg/L are first divided by the molar mass of the substance;
- yields given in C-mol of biomass per N-mol of substrate are used as mol/mol.

The simulations themselves are run with dimensionless variables, obtained from these dimensional values in Chapter 5.

3.2 The parameter groups

The parameters fall into seven groups. This section explains what each group means. The values are given in the sections that follow.

Growth and death rates. The maximal growth rate r_i is the rate at which species i grows when all its substrates are plentiful. The death coefficient b_i multiplies the total density ρ, so crowding increases mortality. In all three presets $b_i = 0.1\, r_i$.

Half-saturation constants. The constant K_{ij} is the concentration of substrate j at which species i reaches half of its maximal growth rate in the Monod function. A small K_{ij} means that the species grows well even when the substrate is scarce.

Yields and production. The yield Y_{ij} is the amount of biomass made per unit of substrate j consumed by species i. It sets how fast a substrate is used up for a given growth rate. The oxygen yields follow from the yields on the nitrogen substrate and the oxygen use of each reaction (Section 2.1). The production fraction β is the share of the consumed nitrogen that is released as product; it equals 1 in all presets.

Substrate diffusion. The coefficient D_j sets how fast substrate j spreads. It enters the time-scale ratio ε of Equation (2.9), so it decides how strongly the substrates are separated from the bacteria in time.

	AOB	NOB	CMX
Growth rate r_i (1/day)	0.5	1.5	0.375
Death coefficient b_i (1/day)	0.05	0.15	0.0375
K for the nitrogen substrate (μM)	1 (NH_4^+)	1 (NO_2^-)	1 (NH_4^+)
K for O_2 (μM)	3.13	3.13	3.13
Yield on the nitrogen substrate	0.0409	0.0242	0.0651
Yield on O_2	0.06135	0.0121	0.1302

	NH_4^+	NO_2^-	NO_3^-	O_2
Diffusion coefficient D_j (m^2/day)	$8.64 \cdot 10^{-5}$	$8.64 \cdot 10^{-5}$	$8.64 \cdot 10^{-5}$	$8.64 \cdot 10^{-5}$
Boundary value c_j^∞ (μM)	554	0	0	187.5

Table 3.2: Parameters of the preset `rebeca`. Motion $d_i = 10^{-6}$ m^2/day, $a_i = 10\, d_i$, production fraction $\beta = 1$ and $L = 500\,\mu$m, as in all presets.

Bacterial movement. The coefficient d_i is the random motion of species i and a_i is the strength of the pressure-driven movement. In all presets $d_i = 10^{-6}\,m^2$/day and $a_i = 10\, d_i$.

Boundary values. The value c_j^∞ is the concentration of substrate j outside the aggregate. Ammonium and oxygen are supplied from outside, so their values are positive. Nitrite and nitrate are only products, so their outside values are zero.

Domain size. The length L is the radius of the aggregate. All presets use $L = 500\,\mu$m.

3.3 Preset `rebeca`

The preset `rebeca` is a rough estimate that R. Gonzalez-Cabaleiro sent by e-mail, together with a spreadsheet of values. Table 3.2 gives the values as used in the code.

Origin of the values.

- The growth rates are given without a unit in the e-mail. We take them as 1/day.
- The half-saturation constants are 10^{-6} mol/L, which is $1\,\mu$M, for the nitrogen substrates. The oxygen value is $3.13\,\mu$M.
- All substrates have the same diffusion coefficient, 10^{-9} m^2/s, which is $8.64 \cdot 10^{-5}$ m^2/day.
- The boundary values come from feed concentrations of 10 mg/L ammonium (molar mass 18.04 g/mol, giving $554\,\mu$M) and 6 mg/L oxygen (molar mass 32 g/mol, giving $187.5\,\mu$M).

Rescaling of the yields. The e-mail gives the yields as C-mol of biomass per N-mol of substrate, for example 4.09 for AOB on ammonium. We divide all yields by 100, which gives 0.0409. The reason is that 0.0409 is the value of the same yield in the preset `eloi`, and likewise 0.0242 and 0.0651 for NOB and CMX. We therefore assume a slip of units in the e-mail. This is an assumption and has not been confirmed with the source. It matters: it decides whether the simulated aggregate develops an oxygen-free core.

	AOB	NOB	CMX
Growth rate r_i (1/day)	0.9489	0.5030	0.06923
Death coefficient b_i (1/day)	0.09489	0.05030	0.006923
K for the nitrogen substrate (μM)	1.453 (NH_4^+)	$8.894 \cdot 10^{-5}$ (NO_2^-)	0.063 (NH_4^+)
K for O_2 (μM)	3.606	2.594	3.13
Yield on the nitrogen substrate	0.0409	0.0242	0.0651
Yield on O_2	0.02727	0.0484	0.03255

	NH_4^+	NO_2^-	NO_3^-	O_2
Diffusion coefficient D_j (m^2/day)	$1.692 \cdot 10^{-4}$	$1.651 \cdot 10^{-4}$	$1.644 \cdot 10^{-4}$	$1.814 \cdot 10^{-4}$
Boundary value c_j^∞ (μM)	500	0	0	93.8

Table 3.3: Parameters of the preset `eloi`. Motion $d_i = 10^{-6}$ m^2/day, $a_i = 10\, d_i$, production fraction $\beta = 1$ and $L = 500\,\mu$m, as in all presets.

Oxygen yields and the stoichiometry. AOB, NOB and CMX use $\theta = 1.5$, 0.5 and 2 units of oxygen per unit of nitrogen (Section 2.1). A yield is biomass per unit of substrate, so the yield on oxygen must equal the yield on nitrogen divided by θ. The preset `eloi` does this. In `rebeca` the yield on oxygen equals the yield on nitrogen multiplied by θ: $0.0409 \cdot 1.5 = 0.06135$ for AOB, and the same pattern for NOB and CMX. As a result, the oxygen use of each species is wrong by the factor θ^2: too low by 2.25 for AOB and by 4 for CMX, and too high by 4 for NOB. Because of this, the oxygen balance of Chapter 6 does not hold for this preset. The source should be asked whether the oxygen yields were meant in this way.

3.4 Preset `eloi`

The preset `eloi` uses kinetic parameters derived from thermodynamics by Martinez-Rabert et al. [7, 8], at 25°C and pH 7, and literature values for the diffusion coefficients. Table 3.3 gives the values as used in the code.

Origin of the values.

- The growth rates and half-saturation constants come from the supplementary table of [7, 8]. The growth rates are given per hour and are multiplied by 24.
- The yields on the nitrogen substrates come from the same table. It gives no yields on oxygen, so these are derived from the stoichiometry: the yield on oxygen is the yield on nitrogen divided by the oxygen use per nitrogen (1.5, 0.5, 2).
- The diffusion coefficients are literature values in m^2/h, multiplied by 24.
- The kinetic data contain no feed concentrations. The boundary values 500 μM for ammonium and 93.8 μM for oxygen (about 3 mg/L) are choices made in the code.

[TODO: Confirm the source of the feed concentrations 500 and 93.8 μM.]

Parameter	Case A	Case B	Case C
$d_1 = d_2\,(= d_3)$	10^{-6}	10^{-6}	10^{-6}
$a_1 = a_2\,(= a_3)$	10^{-5}	10^{-5}	10^{-5}
r_1	1	0.5	1
r_2	1	1	1
r_3	1	—	1
K_1	1	100	1
K_2	1	100	1
K_3	1	—	1
$b_1 = b_2\,(= b_3)$	0.1	0	0.1
D	10^{-4}	10^{-5}	—
$D_1 = D_2 = D_3$	—	—	10^{-5}
Y_1	0.2	4	10
Y_2	0.2	0.5	10
Y_3	0.2	—	10
σ_1, σ_2	—	—	0.5
L	1	1	1
initial substrate c^0 (A, B); c_1^0 (C)	5	5	5
boundary flux c^∞ (A, B); c_1^∞ (C)	10^{-5}	1	0

Table 3.4: Dimensionless parameters of the three cases of the source paper (Table 1 of [3]). A dash means not applicable.

Two NOB variants. The table uses the NOB kinetics of *Nitrospira*. The code also stores an alternative set for *Nitrobacter*: $r = 0.7986$ 1/day, $K_{\mathrm{NO_2}} = 0.004447\,\mu$M, $K_{\mathrm{O_2}} = 7.813\,\mu$M, with the same yields. The default is *Nitrospira*.

A very small half-saturation constant. The constant of NOB for nitrite is $8.9 \cdot 10^{-5}\,\mu$M, many orders of magnitude below the other constants. NOB is therefore almost saturated at any nitrite concentration above zero. This makes the reaction terms very sharp near zero nitrite, which matters for the numerical solver.

3.5 The parameter sets of the source paper

The paper gives its parameters in dimensionless form (Table 3.4) for three simulation cases [3]. The domain is the unit square, $L = 1$.

- **Case A:** three identical competing species with one substrate. The species have no advantage over each other.
- **Case B:** two competing species with different strategies. Species 1 is the yield strategist (low r, high Y) and species 2 is the growth strategist (high r, low Y). There is no death ($b = 0$).
- **Case C:** commensalism of three species along a chain of three substrates.

3.6 Parameter sets used in the simulations

This section lists the values used in the numerical experiments. They combine kinetics with choices for the boundary conditions, the initial biomass and the numerical settings.

3.6.1 Competition and commensalism runs

These runs follow Cases A and C of the paper, with the following changes:

- **Higher substrate diffusivity.** The substrate diffusion coefficient is $D = 0.2$, instead of 10^{-4} (Case A) and 10^{-5} (Case C). With the paper's values, $\varepsilon = r\,L^2/D$ is about 10^4, which is far from the regime $\varepsilon \ll 1$ in which the quasi-steady reduction is valid.
- **Consumption strength.** The dimensionless consumption strength $\Lambda = r/(D\,Y)$ (Chapter 5) falls from $5 \cdot 10^4$ to 25 in Case A and from 10^4 to 0.5 in Case C, by factors of 2000 and 20 000. This is the main departure from the paper.
- **Case A with more substrates.** The competition runs add nitrite and nitrate to the single substrate of the paper, and use oxygen as a non-limiting substrate ($K_{O_2} = 10^{-8}$).

Boundary conditions. Three regimes are applied to the substrates that are fed from outside (ammonium and nitrite in competition, c_1 in commensalism):

Regime	Specification
Dirichlet	fixed value 5
Neumann	flux equal to the paper's 10^{-5} divided by D: $-5 \cdot 10^{-5}$ in the quasi-steady runs
Robin	Bi = 5, outside value 5

Oxygen has a Dirichlet value of 5 and nitrate a Dirichlet value of 0. In commensalism, c_2 and c_3 have zero flux.

Initial biomass. Initially, 300 small circles of radius 0.05 are placed at random inside a disc of radius 0.2 at the centre of the domain. Each circle is assigned at random to one of the three species. The total density is normalised to 1 wherever biomass is present (random seed 42). The initial substrate concentration equals the boundary value.

Numerical settings. The grid has 100×100 nodes on the unit square.

Solver	Steps	Time step	Λ (competition / commensalism)	ε
quasi-steady	50	1.0	25 / 0.5	0
coupled	500	0.1	125 / 2.5	25

Snapshots are taken at $t = 0, 5, \ldots, 50$. In the coupled solver $\Lambda = \varepsilon/Y$, so changing ε also changes Λ. An additional scan over $\varepsilon = 1, 5, 20, 25, 50, 100$ uses a 50×50 grid and $T = 10$.

	`rebeca`	`eloi`
Growth rates AOB / NOB / CMX (1/day)	0.5 / 1.5 / 0.375	0.95 / 0.50 / 0.069
K of NOB for nitrite (μM)	1	$8.9 \cdot 10^{-5}$
Diffusion coefficients (m^2/day)	$8.64 \cdot 10^{-5}$ (all equal)	1.64 to $1.81 \cdot 10^{-4}$
Oxygen at the boundary (μM)	187.5	93.8
Time-scale ratio $\varepsilon = \max_j \varepsilon_j$	$4.3 \cdot 10^{-3}$	$1.4 \cdot 10^{-3}$
Oxygen yields consistent with stoichiometry	no	yes

Table 3.5: Comparison of the presets `rebeca` and `eloi`.

3.6.2 AOB and comammox experiments

These runs use the kinetics of the presets `rebeca` and `eloi` (Tables 3.2 and 3.3) with the following choices:

- **Boundary and initial values.** Dirichlet conditions for all substrates, in dimensionless units: ammonium 1, oxygen 5, nitrite and nitrate 0. The initial substrate fields equal these values.
- **Oxygen value.** The value 5 is chosen because the growth of the biomass no longer depends on the oxygen value above about 2 for `eloi` and about 5 for `rebeca`. Oxygen is then not limiting.
- **Bacterial motion.** $d = 10^{-6}$ and $a = 10^{-5}$, as in Case A of the paper, instead of the values of the presets. In dimensionless form the presets' motion is about $2.7 \cdot 10^6$ (`rebeca`) to $4.2 \cdot 10^6$ (`eloi`) times larger, and it would spread the biomass over the whole domain in one step.
- **Survival cutoff.** After every step, densities below 10^{-6} are set to zero. Without this, diffusion places biomass of order 10^{-40} everywhere, and it grows exponentially in the substrate-rich region outside the colony.
- **Initial biomass.** The same 300 circles as above. In the AOB system each circle is AOB or NOB (half each). In the comammox system every circle is CMX. The two systems start with the same total biomass.
- **Grid and time.** 100×100 nodes, time step 0.5, until $T = 50$ or until the biomass first reaches the boundary.
- **Substrates.** Quasi-steady solver, started from the solution of the previous step.

3.7 Comparison and known issues

Table 3.5 compares the two presets used for simulations.

The following points are open or need care:

- The oxygen yields of `rebeca` do not match the stoichiometry, and its factor of 100 in the yields is unconfirmed (Section 3.3).

- The feed concentrations of `eloi` are choices made in the code, not data (Section 3.4).
- The death coefficient is $b_i = 0.1\, r_i$ in all presets. With plentiful substrates, growth and death then balance at a total density of ten times the reference (packed) density, so the presets allow unrealistically dense biomass.
- The motion of the bacteria in the presets is millions of times larger, in dimensionless form, than in the paper. The simulations therefore use the motion of Case A.
- The substrate diffusivity of the competition and commensalism runs is raised relative to the paper (Section 3.6).
- The motion and survival-cutoff choices of the AOB and comammox experiments are not yet agreed with the supervisor.

Appendix A

Explicit Equations of the Nitrification System

This appendix writes the seven equations of the nitrification system with names instead of indices. They are the general equations (Equations (2.1) and (2.2)) with $n = 3$ species and $m = 4$ substrates.

A.1 Names used

Index	Species	Substrate
1	AOB	NH_4^+
2	NOB	NO_2^-
3	CMX	NO_3^-
4	—	O_2

The densities are u_{AOB}, u_{NOB}, u_{CMX} and the concentrations are $c_{\mathrm{NH_4}}$, $c_{\mathrm{NO_2}}$, $c_{\mathrm{NO_3}}$, $c_{\mathrm{O_2}}$. The total density is $\rho = u_{\mathrm{AOB}} + u_{\mathrm{NOB}} + u_{\mathrm{CMX}}$. Parameters carry the same names, for example r_{AOB} for the maximal growth rate of AOB and $Y_{\mathrm{AOB,NH_4}}$ for the yield of AOB on ammonium.

A.2 Growth factors

Each species consumes exactly two substrates, so its growth depends on two Monod factors:

$$M_{\mathrm{AOB}} = \frac{c_{\mathrm{NH_4}}}{K_{\mathrm{AOB,NH_4}} + c_{\mathrm{NH_4}}} \frac{c_{\mathrm{O_2}}}{K_{\mathrm{AOB,O_2}} + c_{\mathrm{O_2}}}, \tag{A.1}$$

$$M_{\mathrm{NOB}} = \frac{c_{\mathrm{NO_2}}}{K_{\mathrm{NOB,NO_2}} + c_{\mathrm{NO_2}}} \frac{c_{\mathrm{O_2}}}{K_{\mathrm{NOB,O_2}} + c_{\mathrm{O_2}}}, \tag{A.2}$$

$$M_{\mathrm{CMX}} = \frac{c_{\mathrm{NH_4}}}{K_{\mathrm{CMX,NH_4}} + c_{\mathrm{NH_4}}} \frac{c_{\mathrm{O_2}}}{K_{\mathrm{CMX,O_2}} + c_{\mathrm{O_2}}}. \tag{A.3}$$

The other two substrates have $K = 0$ for that species, so their factors equal 1 and drop out of the product in Equation (2.1).

A.3 The seven equations

Bacteria.

$$\partial_t u_{\mathrm{AOB}} = \nabla \cdot [d_{\mathrm{AOB}} \nabla u_{\mathrm{AOB}} + a_{\mathrm{AOB}}\, u_{\mathrm{AOB}} \nabla \rho] + u_{\mathrm{AOB}} (r_{\mathrm{AOB}} M_{\mathrm{AOB}} - b_{\mathrm{AOB}} \rho), \tag{A.4}$$

$$\partial_t u_{\mathrm{NOB}} = \nabla \cdot [d_{\mathrm{NOB}} \nabla u_{\mathrm{NOB}} + a_{\mathrm{NOB}}\, u_{\mathrm{NOB}} \nabla \rho] + u_{\mathrm{NOB}} (r_{\mathrm{NOB}} M_{\mathrm{NOB}} - b_{\mathrm{NOB}} \rho), \tag{A.5}$$

$$\partial_t u_{\mathrm{CMX}} = \nabla \cdot [d_{\mathrm{CMX}} \nabla u_{\mathrm{CMX}} + a_{\mathrm{CMX}}\, u_{\mathrm{CMX}} \nabla \rho] + u_{\mathrm{CMX}} (r_{\mathrm{CMX}} M_{\mathrm{CMX}} - b_{\mathrm{CMX}} \rho). \tag{A.6}$$

Substrates.

$$\partial_t c_{\mathrm{NH_4}} = D_{\mathrm{NH_4}} \Delta c_{\mathrm{NH_4}} - \frac{r_{\mathrm{AOB}}}{Y_{\mathrm{AOB,NH_4}}}\, u_{\mathrm{AOB}} M_{\mathrm{AOB}} - \frac{r_{\mathrm{CMX}}}{Y_{\mathrm{CMX,NH_4}}}\, u_{\mathrm{CMX}} M_{\mathrm{CMX}}, \tag{A.7}$$

$$\partial_t c_{\mathrm{NO_2}} = D_{\mathrm{NO_2}} \Delta c_{\mathrm{NO_2}} + \sigma_{\mathrm{AOB}\to\mathrm{NO_2}}\, r_{\mathrm{AOB}}\, u_{\mathrm{AOB}} M_{\mathrm{AOB}} - \frac{r_{\mathrm{NOB}}}{Y_{\mathrm{NOB,NO_2}}}\, u_{\mathrm{NOB}} M_{\mathrm{NOB}}, \tag{A.8}$$

$$\partial_t c_{\mathrm{NO_3}} = D_{\mathrm{NO_3}} \Delta c_{\mathrm{NO_3}} + \sigma_{\mathrm{NOB}\to\mathrm{NO_3}}\, r_{\mathrm{NOB}}\, u_{\mathrm{NOB}} M_{\mathrm{NOB}} + \sigma_{\mathrm{CMX}\to\mathrm{NO_3}}\, r_{\mathrm{CMX}}\, u_{\mathrm{CMX}} M_{\mathrm{CMX}}, \tag{A.9}$$

$$\partial_t c_{\mathrm{O_2}} = D_{\mathrm{O_2}} \Delta c_{\mathrm{O_2}} - \frac{r_{\mathrm{AOB}}}{Y_{\mathrm{AOB,O_2}}}\, u_{\mathrm{AOB}} M_{\mathrm{AOB}} - \frac{r_{\mathrm{NOB}}}{Y_{\mathrm{NOB,O_2}}}\, u_{\mathrm{NOB}} M_{\mathrm{NOB}} - \frac{r_{\mathrm{CMX}}}{Y_{\mathrm{CMX,O_2}}}\, u_{\mathrm{CMX}} M_{\mathrm{CMX}}. \tag{A.10}$$

Ammonium is consumed by AOB and CMX. Nitrite is produced by AOB and consumed by NOB. Nitrate is produced by NOB and CMX and never consumed. Oxygen is consumed by all three.

A.4 Initial and boundary conditions

As in the paper, the bacteria have zero flux through the boundary, and ammonium and oxygen are supplied by a constant flux:

$$\partial_\nu u_{\mathrm{AOB}} = \partial_\nu u_{\mathrm{NOB}} = \partial_\nu u_{\mathrm{CMX}} = 0,$$
$$\partial_\nu c_{\mathrm{NH_4}} = \gamma_{\mathrm{NH_4}} > 0, \qquad \partial_\nu c_{\mathrm{O_2}} = \gamma_{\mathrm{O_2}} > 0,$$
$$\partial_\nu c_{\mathrm{NO_2}} = \partial_\nu c_{\mathrm{NO_3}} = 0.$$

Initially $c_{\mathrm{NH_4}}$ and $c_{\mathrm{O_2}}$ are positive constants and $c_{\mathrm{NO_2}} = c_{\mathrm{NO_3}} = 0$. Other boundary conditions are used later in this thesis.

Bibliography

[1] Holger Daims, Elena V. Lebedeva, Petra Pjevac, Ping Han, Craig Herbold, Mads Albertsen, Nico Jehmlich, Marton Palatinszky, Julia Vierheilig, Alexandr Bulaev, Rasmus H. Kirkegaard, Martin von Bergen, Thomas Rattei, Bernd Bendinger, Per H. Nielsen, and Michael Wagner. Complete nitrification by Nitrospira bacteria. *Nature*, 528(7583):504–509, 2015.

[2] Neil Fenichel. Geometric singular perturbation theory for ordinary differential equations. *Journal of Differential Equations*, 31:53–98, 1979.

[3] Viktoria Freingruber, Rebeca Gonzalez-Cabaleiro, and Havva Yoldaş. Ecological interactions and spatial dynamics in microbial aggregates: A novel modelling framework, 2025.

[4] Ferdi L. Hellweger, Robert J. Clegg, James R. Clark, Caroline M. Plugge, and Jan-Ulrich Kreft. Advancing microbial sciences by individual-based modelling. *Nature Reviews Microbiology*, 14(7):461–471, 2016.

[5] Daniel Henry. *Geometric Theory of Semilinear Parabolic Equations*, volume 840 of *Lecture Notes in Mathematics*. Springer, 1981.

[6] Jan-Ulrich Kreft. Biofilms promote altruism. *Microbiology*, 150(8):2751–2760, 2004.

[7] Eloi Martinez-Rabert, Chiel van Amstel, Cindy Smith, William T. Sloan, and Rebeca Gonzalez-Cabaleiro. Environmental and ecological controls of the spatial distribution of microbial populations in aggregates. *PLOS Computational Biology*, 18(12):e1010807, 2022.

[8] Eloi Martinez-Rabert, Cindy J. Smith, William T. Sloan, and Rebeca Gonzalez-Cabaleiro. Competitive and substrate limited environments drive metabolic heterogeneity for comammox Nitrospira. *ISME Communications*, 3(1):91, 2023.

[9] A. N. Tikhonov. Systems of differential equations containing small parameters in the derivatives. *Matematicheskii Sbornik*, 31(73):575–586, 1952.

[10] Maartje A. H. J. van Kessel, Daan R. Speth, Mads Albertsen, Per H. Nielsen, Huub J. M. Op den Camp, Boran Kartal, Mike S. M. Jetten, and Sebastian Lücker. Complete nitrification by a single microorganism. *Nature*, 528(7583):555–559, 2015.

[11] Oskar Wanner, Herman J. Eberl, Eberhard Morgenroth, Daniel R. Noguera, Cristian Picioreanu, Bruce E. Rittmann, and Mark C. M. van Loosdrecht. *Mathematical Modeling of Biofilms*. IWA Scientific and Technical Reports. IWA Publishing, 2006.